

Climate Analysis with LLMs: Examining the Inevitable Destruction of the Planet*

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Abstract—In the context of Technological Foresight, this article investigates the potential of Large Language Models (LLMs) to automate the generation of climate reports, focusing on the projected climate evolution in Brazil over the next 50 years. This study addresses the challenge of rapidly and accurately analyzing extensive climate datasets, utilizing generative LLM to interpret outputs from predictive climate models and generate detailed reports that facilitate strategic decision-making. The proposed approach aims to streamline the integration of predictive insights related to Brazil's climate by automating report generation, thus supporting governmental and organizational efforts with timely and reliable information on future climate scenarios. This project includes the collection of historical climate data specific to Brazil and the development of time series predictive models that estimate temperature and other climate metrics for each decade over the next half-century. Visualizations of Brazil's projected climate trends are also generated, providing clear insights into anticipated changes. The outputs from these predictive models are processed by agents and generative AI systems to produce comprehensive, automated reports. Ultimately, this study contributes to the advancement of automated climate analysis, enabling informed decisions driven by AI and expediting the generation of strategic climate reports.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Climate change has emerged as one of the most pressing global challenges of the 21st century, with profound impacts on ecosystems, economies, and societies. Concern for climate analysis, however, is not a recent phenomenon. As early as the 19th century, Howard [1] conducted a series of experiments and observations on the impacts of human activities on the climate, focusing specifically on London. Despite its geographic limitations, Howard's work demonstrates that both concern for climate change and the necessity to study it were already relevant issues at that time. Over the decades, climate science has evolved significantly, becoming a multidisciplinary

field that integrates environmental science, technology, policy, and public communication. Shi et al. [2] understands the importance of effective communication in shaping public perception of climate change, emphasizing culturally sensitive strategies to enhance acceptance of scientific findings and foster collective action. Their work reveals the importance of aligning scientific messaging with diverse cultural perspectives to advance support for climate policies. Recent studies have underscored the critical nature of human-induced climate change and its destructive consequences.

Wuebbles [3] identifies global warming as partly attributable to the increased concentration of greenhouse gases, such as carbon dioxide, resulting from fossil fuel combustion. These gases, although naturally present in the atmosphere, play a very important role in the greenhouse effect by trapping longwave radiation emitted from Earth's surface [4] [5]. This process, while essential for maintaining Earth's habitability, intensifies as greenhouse gas concentrations rise, leading to significant alterations in precipitation patterns, disruptions in the global hydrological cycle, and an increase in the frequency of extreme weather events, such as droughts and floods. Lelieveld [6] brings more visibility to the dual implications of addressing greenhouse gas emissions: while eliminating fossil fuel emissions could prevent millions of deaths annually and improve water security, it might also result in a temporary rise in global temperatures due to aerosol removal. These studies collectively emphasize the urgency of reducing emissions to limit global warming and mitigate its adverse impacts.

In addition to these global phenomena, localized effects, such as the Urban Heat Island (UHI), exacerbate the challenges posed by climate change. The UHI effect arises from factors such as urbanization, reduced vegetation, and the thermal properties of urban materials like concrete and asphalt as seen in [7]–[9]. This effect intensifies temperature disparities

between urban and rural areas, increases energy demands for cooling, and poses significant public health risks, particularly under scenarios of climate change. Furthermore, the agricultural sector, a vital component of many economies, is increasingly vulnerable to the impacts of climate change. Springmann et al. [10] discuss the potential consequences of reduced food availability, particularly for fruits, vegetables, and proteins, on global health outcomes. Specifically, in Brazil, Bunn [11] [11] projects that the cultivation of Arabica coffee, a key agricultural product, could face substantial challenges due to temperature and precipitation changes, threatening the economic and environmental sustainability of this crop.

A. Identified Problem

While significant progress has been made in global climate science, there is a noticeable gap in studies focused specifically on Brazil. Despite being one of the largest contributors to global agriculture and biodiversity, Brazil remains underrepresented in climate forecasting research. This observation is supported by a systematic search of the Scopus database, which revealed approximately 500 articles globally related to Machine Learning (ML) and climate forecasting. However, when refined to include studies specifically addressing Brazil, this number decreases to 83. These findings uncover the critical need for a systematic study tailored to Brazil's climatic and environmental conditions.

B. Motivation and Objectives

All the studies presented so far serve primarily as a contextual foundation to demonstrate the importance of research on climate change. However, these studies do not represent the primary motivation for this specific research. The motivation for this study arises from the urgent need to address the identified research gap and to develop advanced tools for long-term climate forecasting specifically tailored to the Brazilian context. While climate data analysis is a widely researched topic on a global scale, the scarcity of region-specific studies focused on Brazil represents a significant limitation in effectively addressing the challenges posed by climate change in this region. This research seeks to bridge that gap by exploring state-of-the-art methodologies and technologies to provide actionable insights and targeted forecasts for Brazil. By addressing the distinctive climatic and environmental conditions of the country, this study not only contributes to a better understanding of the local impacts of climate change but also supports the formulation of strategies for mitigation and adaptation tailored to Brazil's specific needs.

The objective of this study is to develop an automated system that combines data collection, ML models, and LLM - based analysis to provide actionable insights for climate forecasting in Brazil over the next 30 years. The proposed approach involves three stages: (1) collecting and preprocessing historical data on climate-related variables in Brazil; (2) training ML models to generate predictions; and (3) analyzing the model outputs using LLM - based agents to interpret results and provide decision-making insights. These three

stages are designed to provide a comprehensive and systematic framework for understanding climate trends and their potential impacts in Brazil.

C. Importance of Weather Forecasting in Brazil

Predicting temperature, humidity, and other climatic conditions in Brazil is crucial due to the region's unique environmental and economic challenges under climate change. [12] Brazil's agribusiness sector, a global leader in soybean, coffee, and sugarcane production, is highly sensitive to climatic variability. Accurate forecasts can optimize planting, harvesting, and resource allocation, reducing crop failures and environmental impacts. Rising temperatures and reduced humidity also exacerbate wildfires, particularly in the Amazon and Cerrado, where predictions can enhance prevention and response efforts. Additionally, water resources critical for hydroelectric power, agriculture, and urban supply are directly influenced by climate variability, making forecasting essential for sustainable management. In urban areas, accurate predictions can guide adaptive infrastructure planning to mitigate flooding, heatwaves, and air quality issues, protecting public health and reducing inequalities. Forecasting Brazil's climate conditions is essential to support agriculture, biodiversity, and urban resilience in the face of climate change.

D. Research Proposal

Given all of these reasons as a motivation, this paper proposes a multi-stage project to conduct climate forecasting for Brazil over the next 30 years, with forecasts provided in 10-year intervals. The project employs a three-stage process [Figure 1]:

- **Data Collection and Preprocessing:** Historical climate data, including temperature, precipitation, atmospheric pressure, and humidity, will be gathered and preprocessed to ensure quality and consistency.
- **Machine Learning Models:** Advanced ML techniques will be applied to train predictive models capable of delivering accurate climate forecasts for Brazil.
- **Multi-Agent Analysis with LLMs:** The outputs of the predictive models will be analyzed using LLM-based agents to extract meaningful insights and synthesize actionable information.

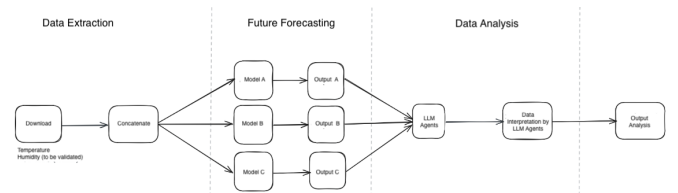


Fig. 1. Solution design

II. RELATED WORKS

Given the relevance of the topic, several studies have explored the theme of climate change, as highlighted in

the introduction. However, the methodology proposed in this work is based on the application of Machine Learning (ML) and LLMs for automated prediction and analysis. Thus, this chapter discusses related studies that share similar methodologies. [13] Eyring et al. explore the application of ML to enhance climate modeling and analysis, emphasizing the use of hybrid models that integrate machine learning techniques with physical approaches to improve the consistency of climate projections. Additionally, the study addresses the use of ML-based emulators to generate quick and robust projections of extreme climate events. The article also discusses challenges such as uncertainty quantification, model explainability (XAI), and causal inference, which are considered essential to improve the reliability of predictions.

[14] Duan et al., in turn, present robust methodologies to predict and analyze climate change using Integrated Assessment Models (IAMs). These models combine multiple perspectives and variables, such as climatic, economic, and energy aspects, resulting in more accurate predictions and reliable insights. The study highlights the importance of multidisciplinary approaches for effective climate policy formulation, with a focus on mitigation and adaptation strategies.

[15] Another relevant work is that of Li et al., which proposes a hybrid ML model called SENB-EM, capable of simultaneously performing climate predictions through classification and regression. This approach reflects efforts to integrate models to improve prediction accuracy. The model also employs a voting system to reduce errors, which resembles the automation of insights explored in this study. Furthermore, the article applies its analyses to specific variables, such as Sea Surface Temperature (SST) indices, demonstrating its effectiveness in forecasting atmospheric patterns.

[16] Chantry et al. investigate the use of ML in climate and meteorological modeling, addressing its applications at different temporal scales, from short-term forecasts to long-term projections. The study emphasizes the role of machine learning in bias correction, downscaling, and physical parameterization, areas that reflect the objectives of automated analysis in this work. The integration of traditional methods with ML is also highlighted as a way to improve computational efficiency and prediction accuracy.

[17] Zepner et al. introduce an interactive platform for visualizing historical and current climate data. This approach aims to facilitate access to climate information for diverse audiences by transforming large data volumes into comprehensible and useful representations. Although the study focuses on presenting geospatial patterns and climate variables in interactive platforms, it differs from the present work by prioritizing data accessibility rather than forecasting specific variables such as temperature, relative humidity, and atmospheric pressure.

[18] ROLNICK et al. explores the integration of machine learning (ML) techniques in addressing climate change through mitigation and adaptation strategies. It provides an analysis of the applications of ML in several domains. In electricity systems, ML contributes to enhancing renewable energy integration by improving forecasting accuracy for solar

and wind power, optimizing grid operations, and accelerating advancements in energy storage technologies. In the transportation sector, ML aids in reducing greenhouse gas emissions by optimizing vehicle routing, improving fuel efficiency, supporting electric vehicle (EV) technologies, and enabling freight consolidation. For buildings and urban planning, ML facilitates energy optimization through smart systems, enhances the accuracy of energy demand modeling, and supports the development of energy-efficient urban infrastructures. The article also shed a light to the use of ML in carbon dioxide removal technologies, precision agriculture, and the monitoring of natural ecosystems to reduce emissions and enhance environmental resilience. Advanced ML methods, including reinforcement learning and deep learning, are applied to optimize processes, forecast climate impacts, and design sustainable energy systems. The authors emphasize the importance of data-driven approaches, interdisciplinary collaboration, and the integration of domain-specific knowledge to maximize the efficacy of ML applications in combating climate change.

[19] ARDABILI et al. systematically reviews the application of machine learning (ML) and deep learning (DL) techniques in hydrological processes, climate change, and Earth systems modeling. Its purpose is to evaluate the current state of ML and DL methods in analyzing, predicting, and managing hydrological and climatic phenomena. The authors classify ML methods into categories such as tree-based, support vector machines, neural network-based, and hybrid techniques, while also examining the utility of DL models like convolutional neural networks (CNNs), recurrent neural networks (RNNs), and other advanced architectures in addressing complex hydrological challenges. The study concludes that while ML methods are widely established and demonstrate significant efficiency in hydrological predictions, DL approaches are still in developmental stages with ongoing research. Novel methods with higher accuracy and robustness, such as hybrid and ensemble techniques, are emerging as powerful tools for climate modeling. The authors emphasize the potential of these technologies in reducing computation costs, enhancing robustness, and increasing accuracy in predictions for environmental processes.

[20] O'GORMAN et al. propose the use of machine learning, specifically Random Forests (RF), as a parameterization method for moist convection within general circulation models (GCMs). The goal is to address uncertainties and improve the simulation of climate statistics, precipitation extremes, and climate change by replacing conventional physical parameterizations with ML-based ones. The study demonstrates that the RF parameterization can accurately simulate the mean climate and precipitation extremes without specialized training. It shows that ML models trained on data from both current and warmer climates can generalize effectively to simulate climate change. Even when trained solely on the warmer climate, the model performed well due to overlaps in temperature and humidity patterns between latitudes of different climates. This approach successfully replicated temperature profiles, precipitation distributions, and the effects of global warming

in a controlled simulation setting. The authors came to the conclusion that ML-based parameterizations, when properly trained, can not only enhance the robustness of climate simulations but also offer new tools for analyzing the interactions between convection and large-scale atmospheric conditions. This aligns with the goal of utilizing machine learning to make accurate predictions about climate change, focusing on improved modeling of critical variables like precipitation and atmospheric dynamics under changing climatic conditions.

[21] CRANE-DROESCH et al. presents an innovative approach to modeling the relationship between crop yields and weather conditions using a semiparametric neural network (SNN). This method integrates parametric statistical models with deep neural networks, allowing it to capture complex, nonlinear relationships in high-dimensional datasets while leveraging known parametric structures and addressing unobserved heterogeneity. The SNN approach outperforms classical statistical methods and fully nonparametric neural networks in predicting crop yields under various climate scenarios. By using a hybrid model, the SNN achieves higher predictive accuracy and precision, particularly for extreme weather conditions that significantly impact yields. The study applies this methodology to corn yield data from the U.S. Midwest, showing that projections of yield decline due to climate change are less severe compared to traditional methods. This is attributed to the model's ability to represent the nuanced effects of temperature, precipitation, and other weather variables on crop growth.

[22] DAVENPORT et al. investigates how CNNs can identify and analyze atmospheric circulation patterns associated with extreme precipitation in the U.S. Midwest. By training the CNN on sea-level pressure and 500-hPa geopotential height anomalies, the study demonstrates its effectiveness in predicting extreme precipitation days with 91% accuracy. The findings reveal that, over the past decades, the frequency of extreme precipitation circulation patterns (EPCPs) has remained relatively stable, but precipitation intensity during these patterns has significantly increased. This intensification is attributed to enhanced moisture flux from the Gulf of Mexico, linked to thermodynamic changes rather than dynamic alterations in atmospheric circulation. CNN's interpretability, achieved through visualization techniques like layerwise relevance propagation, provided insights into the physical drivers of these patterns, highlighting the model's ability to identify meaningful meteorological features.

III. RESEARCH METHODOLOGY

The objective of this study is to expand research on climate change with a specific focus on Brazil. As highlighted in previous sections, while there are numerous studies globally on this topic, research centered on Brazil remains relatively scarce. This study is structured into three main phases: data collection, predictive model development, and data analysis using multi-agent systems of LLMs.

A. Data Collection

Meteorological data was sourced directly from the Historical Meteorological Data repository of the Instituto Nacional de Meteorologia (INMET) [23]. This repository provides numerous .zip files, each corresponding to a specific year from 2000 to the present (2024 at the time of this writing). To facilitate efficient data retrieval, a specialized script was developed to automate the process of collecting historical meteorological data through web scraping techniques.

The created script is designed to access the specific INMET webpage where the historical datasets are hosted, identify the available .zip files, and download them to a predefined local directory. This automation was essential to manage the extensive amount of historical data efficiently and avoid the time-consuming process of manual data collection. Automating this task is particularly critical in scientific contexts where large datasets are frequently required for modeling, climate analysis, or related studies. Each .zip file contains several .csv files, with each file representing meteorological data for a specific city in Brazil for an entire year, recorded on an hourly basis. These CSV files include the following key variables:

The names of the variables are in Brazilian Portuguese due to the fact that this is how they are named in the official Meteorological Institute Government website. In addition to these variables, the files also include metadata providing contextual information such as the city, state, altitude, latitude, and longitude of the meteorological station. These data are critical for subsequent analyses, enabling predictive modeling and the identification of climate patterns across Brazil's diverse regions.

B. Predictive Model Development

Prediction models represent an intermediate layer in the overall process. Using the time series data retrieved on previous step (data collection), the goal is to forecast future values. A proof of concept was conducted to predict a specific feature (temperature) based on multiple historical data features, with data ranging from 2000 to 2024, as detailed in Section III.a. The development of prediction models involved several stage, such as: data preprocessing, feature engineering, model development, training and validation, model evaluation, and result presentation.

C. Data Preprocessing

Initial preprocessing was essential to cleanse the dataset. Key procedures included the removal of outliers, handling of missing values, and correction of invalid entries.

- **Outliers** According to Lai et al. [24], in the context of time series, an outlier is defined as an instance or subsequence of data that significantly deviates from the expected behavior or the predominant pattern in the series. Therefore, it is crucial to eliminate outliers. The inter quartile range (IQR) method was utilized for this purpose.

Vinutha et al. [25] describe the inter quartile range as the difference between the first quartile (Q1) and the third

Column Name	Data Portuguese Name
DATE (YYYY-MM-DD)	DATA (YYYY-MM-DD)
TIME (UTC)	HORA (UTC)
TOTAL PRECIPITATION, HOURLY (mm)	PRECIPITAÇÃO TOTAL, HORÁRIO (mm)
ATMOSPHERIC PRESSURE AT STATION LEVEL, HOURLY (mB)	PRESSÃO ATMOSFÉRICA AO NÍVEL DA ESTAÇÃO, HORÁRIA (mB)
MAX ATMOSPHERIC PRESSURE PREVIOUS HOUR (mB)	PRESSÃO ATMOSFÉRICA MÁX. NA HORA ANT. (AUT) (mB)
MIN ATMOSPHERIC PRESSURE PREVIOUS HOUR (mB)	PRESSÃO ATMOSFÉRICA MÍN. NA HORA ANT. (AUT) (mB)
GLOBAL RADIATION (KJ/m ²)	RADIAÇÃO GLOBAL (KJ/m ²)
AIR TEMPERATURE - DRY BULB, HOURLY (°C)	TEMPERATURA DO AR - BULBO SECO, HORÁRIA (°C)
DEW POINT TEMPERATURE (°C)	TEMPERATURA DO PONTO DE ORVALHO (°C)
MAX AIR TEMPERATURE PREVIOUS HOUR (°C)	TEMPERATURA MÁXIMA NA HORA ANT. (AUT) (°C)
MIN AIR TEMPERATURE PREVIOUS HOUR (°C)	TEMPERATURA MÍNIMA NA HORA ANT. (AUT) (°C)
MAX DEW POINT TEMPERATURE PREVIOUS HOUR (°C)	TEMPERATURA ORVALHO MÁX. NA HORA ANT. (AUT) (°C)
MIN DEW POINT TEMPERATURE PREVIOUS HOUR (°C)	TEMPERATURA ORVALHO MÍN. NA HORA ANT. (AUT) (°C)
MAX RELATIVE HUMIDITY PREVIOUS HOUR (%)	UMIDADE REL. MÁX. NA HORA ANT. (AUT) (%)
MIN RELATIVE HUMIDITY PREVIOUS HOUR (%)	UMIDADE REL. MÍN. NA HORA ANT. (AUT) (%)
RELATIVE HUMIDITY, HOURLY (%)	UMIDADE RELATIVA DO AR, HORÁRIA (%)
WIND DIRECTION HOURLY (°)	VENTO, DIREÇÃO HORÁRIA (°)
MAX WIND GUST (m/s)	VENTO, RAJADA MÁXIMA (m/s)
WIND SPEED HOURLY (m/s)	VENTO, VELOCIDADE HORÁRIA (m/s)

quartile (Q3). Using the IQR, outliers can be identified with the following criteria:

- **Lower-Limit:** Lower Limit: $Q1 - 1.5 \times IQR$
- **Upper Limit:** $Q3 + 1.5 \times IQR$

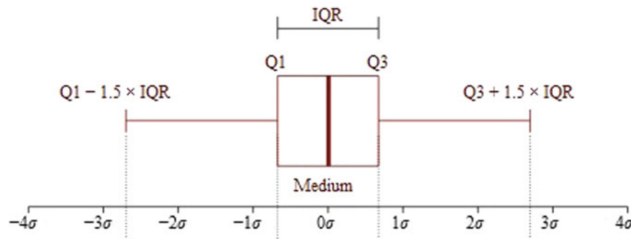


Fig. 2. Representation of Interquartile Range - Box Plot

Any data point below the lower limit or above the upper limit is classified as an outlier as presented in Figure 2.

- **Invalid Values:** Values considered invalid, such as -9999.0, -999.0, -99.0, and 9999.0, were replaced with missing values (NaN).
- **Missing Values:** After replacing invalid values with NaN, the missing values present in the dataset were addressed in two main steps:
 - **Forward Fill:** Initially, the forward fill method was applied to fill data gaps extending up to 6 consecutive hours. This method propagates the last known valid value to fill the missing intervals.
 - **Interpolation:** For more extensive gaps, lasting up to 24 hours, linear interpolation was used for numerical values and the mode for categorical variables.

Both strategies are implemented using the Pandas library [26].

D. Feature Engineering

The feature engineering phase aimed to enhance the predictive performance of the forecasting models by transforming raw meteorological data into informative features. This process started with the extraction of temporal features to capture the inherent seasonal and cyclical patterns present in the meteorological data. Specifically, attributes such as the hour of the day, day of the week, and month were derived from the timestamp of each observation. These temporal features were subsequently converted into cyclical representations using sine and cosine transformations to preserve their periodic nature, enabling the models to effectively recognize and utilize cyclical patterns.

In addition to temporal feature extraction, rolling window statistics were calculated to capture short-term trends and volatility within the data. Rolling means and standard deviations were computed over various window sizes, thereby encapsulating local trends and fluctuations characteristic of weather dynamics. These rolling statistics provide contextual information regarding recent changes in meteorological variables, thereby enhancing the models' capacity to predict future states based on recent behavior.

Furthermore, interaction terms between variables were incorporated to capture complex dependencies that may influence weather dynamics. For example, the interaction between temperature and humidity was modeled to account for phenomena such as variations in the heat index. Including these interaction features allows the models to more accurately represent and predict the multifaceted relationships among different meteorological factors.

To streamline the dataset and enhance computational efficiency, an aggregation process was employed to consolidate

high-frequency hourly meteorological measurements into more manageable daily summaries. This aggregation not only reduced the number of data points but also preserved essential temporal and spatial characteristics necessary for accurate forecasting. The aggregation methodology involved grouping the data by specific categorical and temporal dimensions and applying appropriate statistical functions to numerical and categorical variables.

The aggregation was executed by grouping the data based on multiple dimensions to ensure that the summarized features retained both temporal and spatial context. The aggregation was done by the mean for numerical features and by the mode for categorical features. The primary grouping factors included:

- 1) **Date (Day):** Aggregating data on a daily basis to capture day-level trends and patterns.
- 2) **City:** Consolidating measurements specific to each geographic location to account for spatial variability.
- 3) **State:** Further categorizing data within broader regional boundaries to incorporate state-level climatic influences.
- 4) **Additional Relevant Categories:** Incorporating other pertinent categorical variables present in the dataset, such as weather station identifiers or regional classifications, to maintain comprehensive contextual information.

E. Model Development

A multi-model approach was adopted to leverage the strengths of different forecasting paradigms, ensuring an adequate exploration of the data's patterns from both linear and non-linear perspectives. This strategy integrated classical statistical models, machine learning algorithms, and advanced deep learning architectures. Three distinct models were utilized:

- **Seasonal Autoregressive Integrated Moving Average (SARIMA):** SARIMA models were employed to capture linear temporal dependencies and seasonal effects present in the meteorological data. These models are well-suited for time-series data with clear seasonal patterns, allowing for the modeling of both trend and seasonality components through autoregressive and moving average terms. [27] Dubey et al. explains that SARIMA is used when the data exhibits seasonality, which the standard ARIMA model cannot adequately capture. The process involves transforming the data into a stationary series through differencing (including seasonal differencing) and then fitting the autoregressive and moving average components to capture trends and patterns over time.
- **Long Short-Term Memory (LSTM):** LSTM is a type of recurrent neural network (RNN) utilized to model non-linear relationships and long-term dependencies within the data. LSTMs are adept at handling sequences with temporal dependencies, making them ideal for capturing the intricate patterns in weather data that evolve over extended periods. [28] Yong Yu explains that LSTM is a type of memory cell designed to handle long-term dependencies in sequential data. It overcomes the limitations

of traditional Recurrent Neural Networks (RNNs) by incorporating control mechanisms called "gates," which regulate the flow of information into and out of the memory cell.

- **Temporal Fusion Transformer (TFT):** TFT was integrated to exploit attention mechanisms and multi-horizon forecasting capabilities. TFT models combine the strengths of transformer-based attention mechanisms with recurrent layers, enabling the capture of both short-term and long-term dependencies in the data. This hybrid approach facilitates the modeling of complex temporal dynamics and enhances forecasting performance across multiple future time steps. [29] Lim explains that TFT is designed to forecast variables across multiple time horizons, utilizing attention mechanisms and specialized components that handle heterogeneous and complex data, such as static metadata, past observed inputs, and known future inputs. It also incorporates self-explanatory attention layers to capture long-term relationships and variable selection mechanisms to highlight the most relevant data at each moment.

By employing this ensemble of models, the study ensures that a diverse set of modeling techniques is applied, each contributing unique insights into the data's temporal and spatial structures. This diversity not only improves overall predictive performance but also provides a more resilient forecasting system capable of adapting to various climatic conditions. For all 3 models, additional data treatment was applied such as remissive null values from previous feature engineering, decreasing the precision of the dataset to decrease its size, categorical values treatment and much more. Every step can be consulted in <https://github.com/geeklicantropo/Weather-Forecast-and-Analysis-JOJO>

F. Training and Validation

Each model went through a comprehensive training process, employing cross-validation techniques to evaluate generalization and prevent overfitting. The dataset was divided into training, validation, and testing subsets while preserving temporal order, thereby maintaining the integrity of the time-series structure. This means that the test data was after the training data in temporal order. This temporal partitioning ensures that models are trained on historical data and validated on future data, consistent with real-world forecasting scenarios.

- **Cross-Validation:** Time series cross-validation was implemented using a rolling window approach. This method involves iteratively training the model on an expanding training set and validating it on a fixed-size validation set. It provides reliable estimates of the model's ability to generalize to unseen data and adapt to changing weather patterns.
- **Hyper parameter Tuning:** Hyper parameter optimization was conducted using grid search and randomized search methodologies. Parameters such as learning rates, batch sizes, number of hidden layers, and units were systematically varied to identify the optimal configuration

that minimizes performance metrics on the validation set. This tuning process ensures each model operates efficiently, balancing complexity and computational feasibility.

- **Early Stopping:** Early stopping criteria were applied to prevent overfitting by monitoring validation loss during training. And to prevent an absurd amount of training time. Training was halted when improvements in validation loss plateaued for a predefined number of epochs, ensuring that models do not memorize the training data but instead learn generalizable patterns.

Through training and validation, the study ensures that the predictive models are both accurate and robust, maintaining performance across different temporal segments of the data.

G. LLM Agents

The goal of this part is to build an assistant capable of extracting meaningful insights from climate data, much like a weather expert would do. To achieve this, we merged the historical to present-day temperature data (from 2020 to 2024), along with forecaste data (from 2025 to 2050). The agents should be able to detect patterns in the data and provide a well-structured analysis autonomously - without any human intervention. This could help meteorologists to pos-analyse a big amount of data quickly with agents' help. To enable this, we had to implement several development layers.

The primary challenge involved managing the model's token limits. We used three base Large Language Models: gemma2-9b-it, llama-3.3-70b-versatile, and mixtral-8x7b-32768. Each model supports a different context window: 8,192 tokens, 128,000 tokens, and 32,768 tokens, respectively.

Initially, we considered working with the data's most granular information: Hourly temperature records for each of the 365 days in a year, spanning from 2000 to 2024. This amounts to approximately 24 years of data, which is about $24 \times 365 \times 24 = 219,000$ records. With each record containing around 26 to 32 tokens, we would be dealing with more than 5 million tokens - far exceeding the token context limits of our models.

Realizing this was impractical, we moved to a daily aggregation approach, taking one median data point per day from 2000 through 2050. Over roughly 51 years, we would still surpass 500,000 tokens, which remains beyond a single prompt capability.

Ultimately, we settled on using monthly averages. For the historical period from 2000 to 2024, we have 12 monthly values per year, totaling 288 records. Including the forecast period from 2025 to 2050, that brings the total to about 600 records. We ended up with roughly 19,200 tokens - an amount much more manageable with the LLMs' context window.

Furthermore, we needed multiple agents such as the heat wave expert, cold wave expert and temperature analyzer to read the data simultaneously. Handling all of this in a single prompt was nearly impossible. That's why we decided to use the structural approach described above.

By implementing this structured approach, we can produce a large volume of detailed data without the need for suppression.

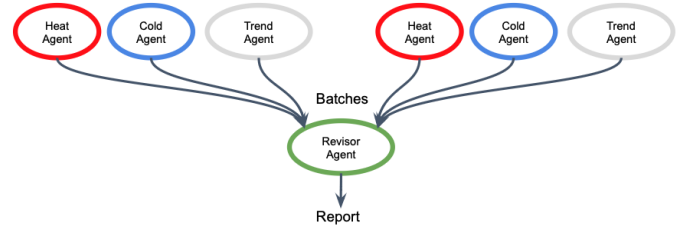


Fig. 3. Agents Algorithm Structure

Each batch or 200 lines is subjected to a rigorous review by specialized agents that identify heat waved, cold waves, and temperature trends as presented in Figure 3. After theses analysis has been completed, all the data is forwarded to a master climate reviewer agent, who then aggregates, synthesizes, and highlights the most important insights, meaning delivering the discover information as a predefined and well-structured set of results.

IV. RESULTS

A. Data Extraction

The data extraction process was completed successfully; however, it was time-consuming due to the large volume of records resulting from the concatenation of 24 years of meteorological data from Brazil. This task had to be executed in batches, with each batch being cleansed and compressed to fit into memory. Additionally, data precision was reduced from float64 and int64 to float32 and int32, and the data aggregation was shifted from hourly to daily intervals, representing the most significant modification to the dataset. Outliers were entirely removed as shown in Figure 4, although the classification of a value as an outlier can be subjective. According to Blázquez-García et al. [30], the determination of outliers depends on the analyst's objectives and the specific aspects being analyzed. In certain contexts, anomalies may be of particular importance. In short, anomalies might be a wanted information that you shouldn't remove.

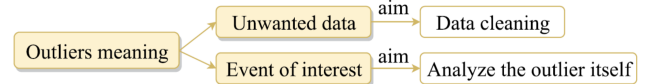


Fig. 4. A value can be an outlier or not depending on the situation

All of those changes made over 70 billions of rows and turned it into 3.5 millions of rows.

B. Data Treatment and Modeling Challenges

As outlined in Section IV.a, several issues were encountered with the available data during the preprocessing phase for developing predictive models. The primary challenges included an excessively large dataset, invalid data entries (e.g., temperatures below -9999 degrees), missing values, and extreme outliers that significantly deviated from the mean. The sheer volume of data presented substantial difficulties in training

models, as the dataset could not be accommodated entirely in memory. To address these limitations, alternative strategies such as batch processing, the utilization of Dask [31] to allow parallel computation across multiple workers in a cluster, and precision reduction were implemented. Approximately 90% of the effort was dedicated to data preprocessing and modeling for the creation of time series predictive models.

C. Data Output Files

Following the data treatment and modeling processes, four types of files were generated:

- **Five-Year Data Files:** Data segmented into five-year intervals starting from 2000, resulting in the following periods:
 - 2000–2004
 - 2005–2009
 - 2010–2014
 - 2015–2019
 - 2020–2024
- **Ten-Year Data Files:** Data segmented into ten-year intervals starting from 2000, resulting in the following periods:
 - 2000–2009
 - 2010–2019
 - 2020–2024
- **Comprehensive Data File:** A single file containing all post-processed data from 2000 to 2024.
- **Sample Data File:** A small subset comprising 100 randomly selected rows for each city, state, and date.

D. Adaptation Due to Technical Constraints

The initial plan was to utilize the entire dataset from the year 2000 onward for developing predictive models. However, technical challenges, particularly related to the dataset's size even after preprocessing, necessitated a shift in strategy. Consequently, a smaller sample was created and employed directly with the agents. This sample serves as input for data analysis and interpretation by the agents. The subsequent section will delve into the creation of these agents and discuss the results obtained.

E. Interpreting Results with Agents

The agent's input data is presented below.

As observed in Figure 5, the raw dataset contains a clear upward trend in temperature values over the years. Naturally, this trend is not perfectly smooth, with a lot of variation and noise. To address this and extract more meaningful patterns, we applied a moving average technique just to display this information - agents only receive the raw data as stated in Figure 6. The graph illustrates a clear upward trend in average annual temperatures, both historical and forecaste temperature, blue and red respectively.

This histogram in Figure 7 shows the distribution of average temperatures, emphasizing that most measurements cluster at higher temperatures as time progresses.

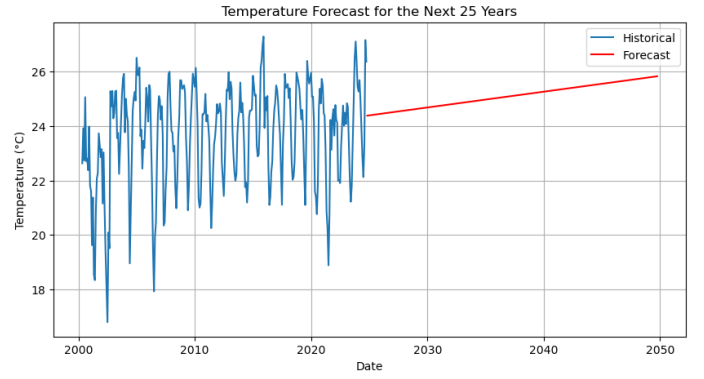


Fig. 5. Historical Temperature from 2000 to 2050

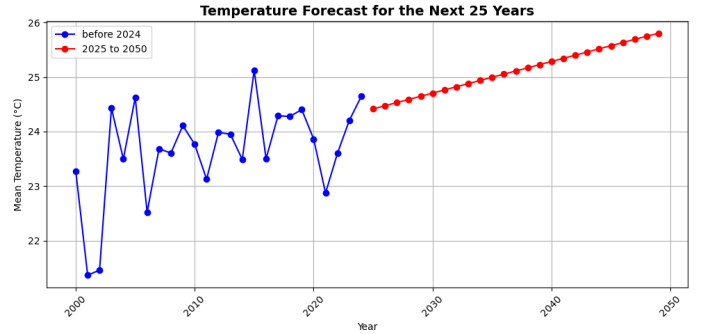


Fig. 6. Historical Moving Average Temperature from 2000 to 2050

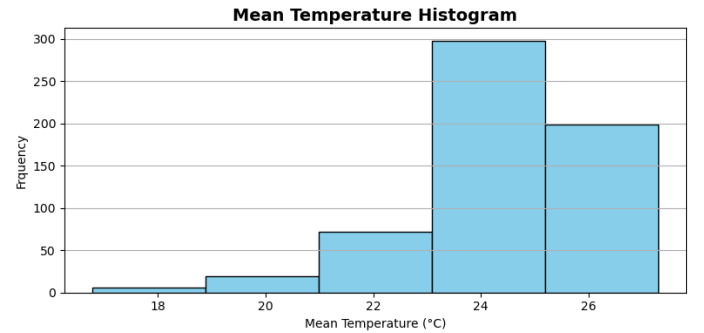


Fig. 7. Temperature Histogram from 2000 to 2050

1) **Heat Waves:** Heat waves could be well defined by region-specific thresholds and prolonged periods of elevated temperatures. This study provided a well fundamental exploration of heat waves in Brazil territory for 50 years. Preliminary evidence suggests notable events in September 2019, 2023, 2024, and 2029, with an average temperature of about 24.543°C. This value is approximately 2.543°C above the baseline for September from all years, indicating a substantial anomaly. However, to confirm the presence and severity of these heat waves, the historical dataset should be standardized according to climatological norms.

2) **Cold Fronts:** In contrast, identifying cold fronts from given data is more challenging, as they depend on a range

of meteorological variables beyond temperature alone. Atmospheric pressure, wind speed and direction, and relative humidity are all critical to classifying a cold front. Without these additional parameters, any conclusion regarding their occurrence remains uncertain. This agent has therefore recommended collecting more detailed, multi-level atmospheric data to correlate sudden temperature drops more effectively and to enhance the accuracy of detecting cold fronts in the climate records.

3) *Climate Trends and Reviewer Output:* The Climate Trend and Reviewer Agents provide an insightful output from this period. They divided the time series into 6 parts with 8 years each. The result is presented in Table I, which summarizes the temperature variations, coldest, and warmest years for each period.

These segmented analyses underline the persistent warming trend, the increasing frequency of anomalies such as heat waves, and the complexities involved in identifying phenomena like cold fronts. Ultimately, the combination of raw data, moving averages, and predictive modeling provides a comprehensive understanding of the evolving climate conditions.

V. COMPARISON BETWEEN RELATED WORKS RESULTS AND OUR RESULTS

The comparative analysis reveals that while both the related works and this study address critical aspects of climate forecasting through ML methodologies, significant distinctions exist in their scope, focus, and regional emphasis. The related works - in Section 2 - primarily concentrate on specific climate-related phenomena or regions, employing ML models to solve narrowly defined problems. For instance, Davenport et al. utilize CNNs to predict extreme precipitation patterns in the U.S. Midwest, offering high accuracy in identifying localized atmospheric circulation phenomena. Similarly, Crane-Droesch et al. develop a semiparametric neural network (SNN) to model crop yields under extreme weather conditions, focusing on agricultural outcomes rather than comprehensive climate features. Chantry et al. emphasize the application of ML in bias correction and downscaling within meteorological models, tackling technical challenges in climate simulation at varying temporal scales.

These studies are important but they are constrained by their geographic specificity and their focus on isolated variables, such as precipitation, sea surface temperatures, or agricultural yields. In contrast, the present study adopts a broader perspective, seeking to predict overarching climate features, including temperature, humidity, atmospheric pressure, and precipitation, specific to Brazil. This regional focus fills a critical gap in climate forecasting research, addressing the lack of Brazil-specific studies highlighted in Section 1. By exploiting ML and LLMs to analyze and interpret predictions, this study advances the field by integrating domain-specific modeling with sophisticated data interpretation techniques. The proposed multi-stage framework—encompassing data collection, ML-based predictive modeling, and LLM-driven insights—aims

to provide actionable and comprehensive forecasts tailored to Brazil’s unique climatic conditions over the next three decades.

From a methodological standpoint, the proposed approach shares similarities with studies like Eyring et al., which emphasize hybrid models combining physical and statistical methods to improve prediction consistency. However, the reliance on LLMs for interpretative analysis represents a novel addition, enhancing the accessibility and applicability of climate data. Additionally, the systematic segmentation of data into five-year and ten-year intervals for trend analysis and the use of moving averages to discern patterns align with best practices observed in related works while adapting them to the regional context.

In terms of results, the studies reviewed demonstrate success in addressing specific challenges. For instance, Davenport et al. achieve high precision in classifying extreme weather events, while Crane-Droesch et al. effectively mitigate prediction errors in agricultural models. However, these results are localized and do not provide the comprehensive insights into long-term climate dynamics that the present study seeks to achieve. By focusing on Brazil and aiming to predict broader climatic variables, this research provides a more extensive understanding of how climate change may manifest in a critical global region. The findings, such as the identification of persistent warming trends and anomalies like heat waves, are pivotal for formulating targeted adaptation and mitigation strategies.

The present study also overcomes notable challenges in data volume and quality, which are common in climate research. While related works like Ardabili et al. emphasize the need for reducing computational costs and improving robustness, this study employs advanced techniques like batch processing and precision reduction to handle large datasets efficiently. The comprehensive preprocessing pipeline, which consolidates 24 years of meteorological data into manageable formats, ensures the reliability and usability of the predictive models.

In conclusion, while the related works contribute valuable insights into specific aspects of climate change, their narrow scope contrasts with the broader, region-specific approach of the present study. By focusing on Brazil, a country of significant ecological and agricultural importance, and employing advanced ML and LLM techniques, this research addresses a critical gap in climate forecasting literature. The results underscore the utility of integrating predictive modeling with interpretative frameworks to deliver actionable insights, setting a foundation for future studies to expand on this innovative methodology.

VI. RECOMMENDATIONS FOR FUTURE ANALYSES

The meteorological data extracted from the databases on the INMET website contained rows and columns with a significant amount of errors. Fields with records of -9999.9 were found in approximately 30% of the total dataset. Naturally, these values are incapable of representing true measurements of the climatic variables captured by the instruments. However, it is notable that these errors were observed more frequently during

TABLE I
TEMPERATURE TRENDS AND PREDICTIONS (2000-2050)

Period	Important Facts	Coldest Year	Warmest Year
2000 - 2008	There was a pronounced upward trend in temperatures during this period. Both the maximum and minimum values reached their highest recorded levels, resulting in a temperature variation of up to 3.201°C.	2001 (21.05°C)	2007 (24.75°C)
2008 - 2016	A consistent average increase of 0.1°C was detected in the moving average of the years during this period.	2011 (23.05°C)	2015 (25.05°C)
2016 - 2024	A significant thermal variation was observed over the three-year span from 2021 to 2024, with 2022 and 2023 each recording progressively higher average temperatures, culminating in even greater values by 2024.	2021 (22.89°C)	2024 (24.70°C)
2024 - 2033	Upward trend detected by Prediction Model.	2025 (24.5°C)	2033 (24.90°C)
2033 - 2042	Upward trend detected by Prediction Model.	2034 (24.95°C)	2042 (25.35°C)
2042 - 2050	Upward trend detected by Prediction Model.	2042 (25.40°C)	2050 (25.80°C)

the period from 2000 to 2008, which may indicate that new technologies or a new methodology for data collection were implemented from that point onward to mitigate errors.

So, once it's indeed necessary to get an more consistent historical database, at the same time, while future studies aim to obtain less problematic historical data, there is also the challenge of finding alternative sources for such data. This is due to it is uncommon for private companies to freely provide access to historical climatic data.

An important point is since our time for writing this present paper was limited, our model lacks the presence of other variables, only handling with temperature. For future works the inclusion of additional climatic variables such as humidity, precipitation, pressure, and wind is crucial for improving the reliability and accuracy of meteorological analyses. While temperature is often the focal variable in climatic studies, these additional factors provide a more comprehensive understanding of weather patterns, trends and possible insights such as study of heat island effects, agricultural productivity, evapotranspiration index, air quality and many others studies.

During the conception of our paper it was decided only working with Brazil climate data, though the use of regional and global climate models could enhance and generate reliable projections for climate-related studies. Regional models allow for a finer spatial resolution, capturing localized phenomena such as microclimates, which are often overlooked in broader analyses. On the other hand, global climate models provide the advantage of understanding large-scale climatic patterns and interactions across continents. Combining both approaches enables researchers to achieve a balance between local accuracy and global context. Furthermore, integrating historical data with machine learning techniques or advanced simulation algorithms can enhance predictive accuracy. By applying these models, future analyses will be better equipped to project trends, identify anomalies, and offer actionable insights for policymakers and stakeholders in mitigating climate-related

risks.

VII. CONCLUSION

Preliminary analyses indicate a general trend of increasing temperatures over recent decades, aligning with global climate change observations. This warming pattern has significant implications for weather variability, agriculture, and well being. Additionally, specific heatwave events have been identified, but their verification requires further analysis to better understand their frequency, duration, and impacts. Detailed studies integrating observational data with predictive models will be essential for anticipating such extreme events and developing mitigation strategies.

The absence of additional climatic variables, such as wind patterns and pressure systems, complicates the analysis of complex phenomena like the identification of cold fronts. These fronts are critical for regulating temperature and precipitation patterns, yet their accurate detection depends on more comprehensive datasets and improved climate models. Addressing these gaps will enhance the understanding of meteorological systems and their broader impacts.

Finally, complementary studies and improved data are necessary to support robust conclusions and inform effective climate adaptation policies. High-quality datasets, combined with advanced modeling techniques, can provide evidence-based insights for designing strategies to mitigate climate risks and build resilience. Collaboration between researchers, governments, and international organizations will be fundamental in addressing the challenges posed by climate change.

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